(Please write your Enrollment Number) Enrollment No. 0160117202 4

MID-TERM EXAMINATION

(Course Name : B.Tech CSE-AI/ ECE-AI) (Semester 5)

(OCTOBER, 2023) OFF LINE mode, <Subjective>

Subject Code: BAI-301	**Subject: Machine Learning(ML)**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) Explain difference between a hard margin and a soft margin in Support Vector Machine (SVM) .		
	(b)Suppose that we have already trained a classification tree, K-Nearest Neighbors, and logistic regression. When the data is relatively large (e.g., larger than one million points), which one of the techniques is slower in making a prediction?. Explain with reason.		
	(c) Explain the role of data splitting in the context of data preprocessing. Why is it essential to divide data into training and testing sets before model training?		
	(d) What role does standardization play in data preprocessing, and how does it impact the training of machine learning models?		
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	
	(a What are different causes of data issues in Machine Learning? What are fallouts?		
	(b) Enumerate and elucidate various techniques employed in data pre-processing, highlighting the significance of each in the context of machine learning.		
	(c) Describe the role of Exploratory Data Analysis (EDA) in the context of Machine Learning and explain why it is considered a crucial step in the machine learning .		
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	
	(a) Suppose you are building a logistic regression model to determine whether or not a person has diabetes. Predicted probabilities(P(Diabetes)) of for 10 patients are given below		

Patient	Diabetes	P(Diabetes)
A	Yes	0.82
B	No	0.37
C	Yes	0.04
D	No	0.41
E	Yes	0.55
F	No	0.62

G	No	0.20
H	Yes	0.91
I	No	0.74
J	Yes	0.34

Assuming that you chose a cut-off value of 0.4, wherein if a probability is greater than 0.4, you would conclude that the corresponding patient has diabetes. If it is less than or equal to 0.4, then you would conclude that they do not have diabetes.

Calculate Accuracy, Precision, Recall, Specificity and F1 Score based on the model?

(b)A dataset of patients who have tested positive or negative for COVID-19 follows. Their symptoms are cough (C), fever (F), difficulty breathing (B), and tiredness (T).

Table-1

	Cough (C)	Fever (F)	Difficulty breathing (B)	Tiredness (T)	Diagnosis
Patient 1	no	yes	yes	yes	Sick
Patient 2	yes	yes	no	yes	Sick
Patient 3	yes	no	yes	yes	Sick
Patient 4	yes	yes	yes	no	Sick
Patient 5	yes	no	no	yes	Healthy
Patient 6	no	yes	yes	no	Healthy
Patient 7	no	yes	no	no	Healthy
Patient 8	no	no	no	yes	Healthy

Using the Gini impurity index, build a decision tree of height 1 that classifies (Diagnosis) this data. What is the accuracy of this classifier on the dataset?

(c) Consider the simplified data set given below.

X	Y
1	5
2	11
3	14
4	18

Using least square method to find the equation of the regression model fitted on this data (X independent and Y is dependent variable). Calculate R square value of this model.

9th oct.